

Acemoglu & Restrepo (2018)

The Race between Man and Machine — displacement, reinstatement, and the condition under which automation raises wages

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Repository 6 · Week of September 21 – 26, 2026

github.com/Arriola123456/ai-06-restrepo

Version read: NBER Working Paper 22252 as revised in June 2017 (87 pp., the course PDF; title *The Race Between Machine and Man*), pinned by SHA-256 in `lean/`. Page numbers are the PDF's (printed page = PDF page – 2). Backup slides after slide 14.

1 · The paper and the problem: the unit of analysis is the economy

Question. Technology enters as *automation* (machines take over tasks previously done by labor) and as *new tasks* (labor gets new things to do). What does each do to output, wages, the labor share and employment — and can the economy keep a stable labor share when both run at once?

This is the only paper of the term whose agent is the aggregate economy. There is no individual optimising over a menu of AI tools: there is a unit measure of tasks $i \in [N - 1, N]$, competitive firms that pick the cheaper factor for each task, a representative household supplying labor, and a capital stock K . The “problem” is a competitive equilibrium, and the objects that move are I (how far automation has reached) and N (how far the task frontier has moved).

Environment (Section 2.1, p. 8–9). Final good $Y = \tilde{B}(\int_{N-1}^N y(i)^{\frac{\sigma-1}{\sigma}} di)^{\frac{\sigma}{\sigma-1}}$. Tasks $i \leq I$ are technologically automated: capital and labor are perfect substitutes there, $k(i) + \gamma(i)l(i)$; tasks $i > I$ need labor with productivity $\gamma(i)$. **Assumption 1:** γ strictly increasing — labor’s comparative advantage rises with the task index. Preferences $u(C, L)$ with an increasing labor supply $L = L^s(\omega)$, $\omega = W/(RK)$ (eq. 11). **Assumption 2:** homothetic demand ($\eta \rightarrow 0$ or $\zeta = 1$), so $\hat{\sigma} = \sigma(1 - \eta) + \zeta\eta$ is the elasticity between capital and labor. **Assumption 3:** $K < \underline{K}$, i.e. $R > W/\gamma(N)$: new tasks are adopted at once.

1 · The threshold: who does which task (p. 10)

Tasks are priced at minimum unit cost,

$$p(i) = \begin{cases} \min\{R, W/\gamma(i)\}^{1-\eta} & i \leq I \\ (W/\gamma(i))^{1-\eta} & i > I. \end{cases}$$

Since γ is increasing there is a unique $\tilde{I}$ with $\frac{W}{R} = \gamma(\tilde{I})$ (eq. 6): capital is cheaper below it, labor above it. Technology may bind before cost does, so the equilibrium threshold is

$$I^* = \min\{I, \tilde{I}\} : \quad i < I^* \text{ produced with capital,} \quad i > I^* \text{ with labor.}$$

Two regimes run through everything that follows:

- ▶ **technology-constrained**, $I^* = I < \tilde{I}$: firms would automate more if they could;
- ▶ **cost-minimising**, $I^* = \tilde{I} < I$: the frontier is slack; factor prices decide.

Proposition 1 (p. 12). Unique equilibrium; output is a CES aggregate with endogenous shares,

$$Y = \frac{B}{1-\eta} \left[(I^* - N + 1)^{\frac{1}{\sigma}} K^{\frac{\sigma-1}{\sigma}} + \left(\int_{I^*}^N \gamma(i)^{\sigma-1} di \right)^{\frac{1}{\sigma}} L^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}},$$

and ω solves the relative demand (13). **Corollary 1:** with $\sigma = \zeta = 1$, $\gamma \equiv 1$, $Y = \frac{B}{1-\eta} K^{1-N+I^*} L^{N-I^*}$: the labor share is $N - I^*$.

2 · The main result I: Proposition 2, the two forces are polar (p. 11)

Let $\varepsilon_L > 0$ be the labor-supply elasticity and $\Lambda_I = \frac{\gamma(I^*)^{\hat{\sigma}-1}}{\int_{I^*}^N \gamma^{\hat{\sigma}-1}} + \frac{1}{I^* - N + 1} > 0$,

$$\Lambda_N = \frac{\gamma(N)^{\hat{\sigma}-1}}{\int_{I^*}^N \gamma^{\hat{\sigma}-1}} + \frac{1}{I^* - N + 1} > 0.$$

Technology-constrained case ($I^* = I < \tilde{I}$):

$$\frac{d \ln(W/R)}{dI} = -\frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L} < 0, \quad \frac{d \ln(W/R)}{dN} = \frac{\Lambda_N}{\hat{\sigma} + \varepsilon_L} > 0, \quad \frac{d \ln(W/R)}{d \ln K} = \frac{1 + \varepsilon_L}{\hat{\sigma} + \varepsilon_L} > 0.$$

Cost-minimising case ($I^* = \tilde{I} < I$): $d \ln(W/R)/dI = 0$ and $d \ln(W/R)/dN = \Lambda_N/(\sigma_{free} + \varepsilon_L) > 0$ with $\sigma_{free} = \hat{\sigma} + \Lambda_I/\varepsilon_\gamma > \hat{\sigma}$.

In all cases the labor share and employment move with ω : $dL/dN > 0$, and when $I^* = I$, $dL/dI < 0$. So automation is always capital-biased (lowers W/R , s_L , L) and new tasks are always labor-biased (raise all three) — *displacement* against *reinstatement*.

2 · The main result II: Proposition 3, the wage is the ambiguous object (p. 13)

Constrained case, with $W/\gamma(I^*) > R > W/\gamma(N)$:

$$d \ln Y|_{K,L} = \underbrace{\frac{B^{\hat{\sigma}-1}}{1-\hat{\sigma}} \left(\left(\frac{W}{\gamma(I^*)} \right)^{1-\hat{\sigma}} - R^{1-\hat{\sigma}} \right)}_{\text{productivity effect of automation} > 0} dI + \frac{B^{\hat{\sigma}-1}}{1-\hat{\sigma}} \left(R^{1-\hat{\sigma}} - \left(\frac{W}{\gamma(N)} \right)^{1-\hat{\sigma}} \right) dN,$$

$$d \ln W = d \ln Y|_{K,L} + (1-s_L) \left(\frac{\Lambda_N}{\hat{\sigma}+\varepsilon_L} dN - \underbrace{\frac{\Lambda_I}{\hat{\sigma}+\varepsilon_L} dI}_{\text{displacement}} \right), \quad d \ln R = d \ln Y|_{K,L} - s_L \left(\frac{\Lambda_N}{\hat{\sigma}+\varepsilon_L} dN - \frac{\Lambda_I}{\hat{\sigma}+\varepsilon_L} dI \right).$$

New tasks always raise W ; automation always raises R and **raises W iff the productivity effect exceeds the displacement effect**. The productivity effect is the cost saving $W/\gamma(I^*) - R$ on the marginal automated task; the displacement effect is $(1-s_L)\Lambda_I/(\hat{\sigma}+\varepsilon_L)$. Cost-minimising case: automation does nothing to factor prices; new tasks raise W and may lower R .

Conditions, spelled out. Assumptions 1–3; $\hat{\sigma} \neq 1$ in the displayed productivity term (Cobb-Douglas is the limit); K fixed (static); the sign of $d \ln W/dI$ depends on K — next slide.

2 · The trap: “does automation necessarily reduce wages and the labour share?”

	$I \uparrow$, constrained	$I \uparrow$, cost-min.	$N \uparrow$
$W/R, s_L, L$	fall, always (Prop. 2)	unchanged	rise, always
R	rises, always (Prop. 3)	unchanged	may fall
W	ambiguous : productivity vs displacement	unchanged	rises, always
$Y _{K,L}$	rises	unchanged	rises

Flat answer: “yes, automation lowers wages and the labor share.” **Paper:** the labor share, yes, always (when technology binds); the wage, no — it rises when the cost saving on the marginal task is large.

The condition, in closed form (Corollary 1, my derivation, proved in Lean): with $Y = cK^{1-s}L^s$, $s = N - I^*$, $\ln W = \ln c + \ln s + (1 - s) \ln K + (s - 1) \ln L$, so

$$\frac{d \ln W}{d I^*} = \ln \frac{K}{L} - \frac{1}{N - I^*} > 0 \iff \frac{K}{L} > e^{1/(N - I^*)} =: \bar{K}.$$

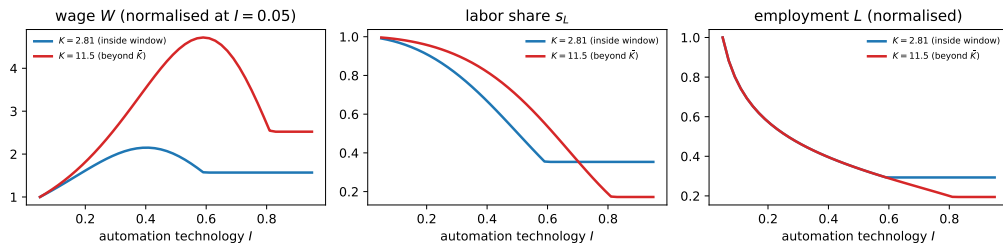
Automation raises the wage when capital is *abundant* enough — R low, the saving $W - R$ large. And $\bar{K}/L = e^{1/s} > \underline{K}/L = (1 - s)/s$: inside Assumption 3 ($K < \underline{K}$) the wage always falls. **Proposition 3 prints the opposite direction** (“increases the wage when $K < \bar{K}$ ”); slide 12.

3 · What I did analytically and computationally

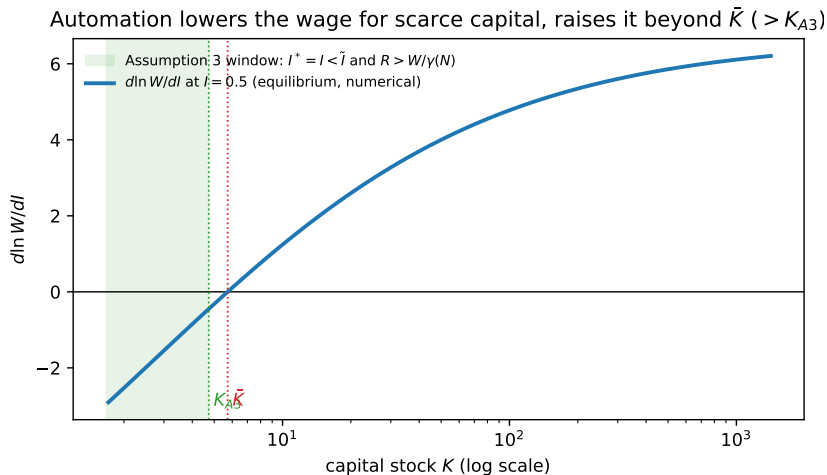
- ▶ **By hand** (hand/, slides 14 and B2): the static model from scratch — the task CES, the unit-cost minimisation behind $p(i)$ in (5), the threshold $W/R = \gamma(\tilde{I})$ and $I^* = \min\{I, \tilde{I}\}$, the two regimes, Assumption 3, and the task demand (7).
- ▶ **Numerics** (analysis/task_model.py): the static equilibrium of Section 2 solved from eq. (13) with $\gamma(i) = 0.3e^{3i}$, $\hat{\sigma} = 0.6$, $\varepsilon_L = 0.3$, $L^s(\omega) = \omega^{\varepsilon_L}$; the two regimes detected from $W/R \gtrless \gamma(I)$; prices from the ideal price index (10).
- ▶ **Check against the paper**: the numerical total derivative $d \ln W / dI$ equals Proposition 3's formula (productivity minus displacement) to four decimals at every point tried.
- ▶ **Findings**: s_L and L fall with I everywhere; W falls throughout the Assumption-3 window ($K < \underline{K} = 4.73$ at $I = 0.5$) and rises only beyond $\bar{K} = 5.73 > \underline{K}$; new tasks raise W and s_L ; the closed-form Cobb-Douglas threshold $\bar{K}/L = e^{1/(N-I^*)}$ matches.
- ▶ **Lean via AppliedModelingLib** (slides 10–12): ten source-facing statements of Section 2 proved; the Cobb-Douglas wage response as an extension. Status partially formalized.

3 · Automation: s_L and L always fall, W may go either way

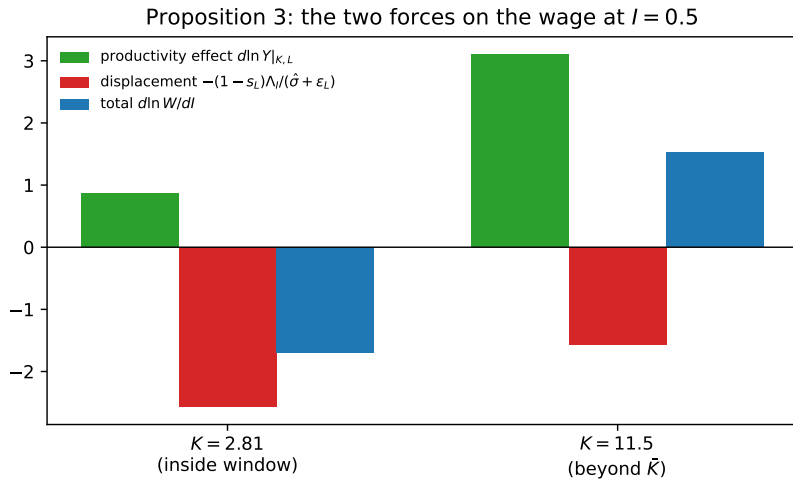
Automation: s_L and L always fall; W falls inside the window, rises beyond $\bar{K}$ ($\hat{\sigma}=0.6$, $\varepsilon_L=0.3$)



3 · $d \ln W / dI$ against K : negative inside Assumption 3, positive beyond $\bar{K}$



3 · Proposition 3 decomposed: productivity effect against displacement effect



4 · Lean I: the run, the statement spec, the status

Workflow (AppliedModelingLib, clone at 2db7d108). Pin the June-2017 PDF by SHA-256 → `init-spec` → statement spec with 10 targets (PDF-page locators, literal source statements, exact Lean types) → `paper_contribution.py new -statement-spec` (the scaffold Lean-validates every Spec) → proofs in `ProofInterface.lean`, extension in `MainTheorems.lean` → `lake build AR18RaceManMachine` → `check AR18RaceManMachine -fast`. **Agent:** Claude Code, not Codex (my decision, four hours before the deadline; disclosed in `lean/docs/RUN_LOG.md`).

The ten Specs (`lean/PaperInterface.lean`), all proved, no sorry: the cost threshold of (5)–(6) with γ strictly increasing; $I^* = \min\{I, \tilde{I}\}$; $\Lambda_I, \Lambda_N > 0$; Prop. 2's three signs in the constrained case and $\sigma_{free} > \hat{\sigma}$; Prop. 3's productivity effect > 0 on both sides of $\hat{\sigma} = 1$; automation raises R ; the wage decomposition as an iff; new tasks raise W ; Corollary 1's marginal products as real derivatives and the labor share $N - I^*$.

Result: `lake build`: Build completed successfully (3313 jobs); `check -fast`: exit 0. **Status: partially formalized.** The equilibrium map (13) and the derivatives of Prop. 2 are taken as the paper's displays; the general $\bar{K}$ is not derived. **Blocker, stated exactly:** the library root `AppliedModelingLib.olean` does not build on my laptop (disk-bound, 12 h last week), so the paper modules import three `Mathlib` modules instead of `import Mathlib`, and the spec validation ran under those imports instead of `import AppliedModelingLib`. No library declaration is used.

4 · Lean II (required slide): the cost threshold of equations (5)–(6)

Paper (p. 10). $p(i) = \min\{R, W/\gamma(i)\}^{1-\eta}$ for $i \leq I$; “there is a unique threshold $\tilde{I}$ such that $W/R = \gamma(\tilde{I})$. For all tasks $i < \tilde{I}$, we have $R < W/\gamma(i)$ ” (Assumption 1: γ strictly increasing).

```
def paper_cost_thresholdSpec : Prop :=
  ∀ (W R : ℝ) (γ : ℝ → ℝ) (i It : ℝ),
    0 < W → 0 < R → StrictMono γ → (∀ x, 0 < γ x) → γ It = W / R →
      (i < It → R < W / γ i) ∧ (It < i → W / γ i < R)

theorem paper_cost_threshold : paper_cost_thresholdSpec := by
  intro W R γ i It hW hR hmono hpos hIt
  have hγi : 0 < γ i := hpos i
  constructor
  · intro hi
    have h1 : γ i < γ It := hmono hi          -- Assumption 1 does the work
    rw [hIt] at h1                             -- γ i < W / R
    rw [lt_div_iff0 hR] at h1                  -- γ i * R < W
    rw [lt_div_iff0 hγi]                       -- goal: R * γ i < W
    linarith [mul_comm R (γ i)]
  · intro hi
    have h1 : γ It < γ i := hmono hi
    rw [hIt] at h1; rw [div_lt_iff0 hR] at h1; rw [div_lt_iff0 hγi]
    linarith [mul_comm R (γ i)]
```

Reading. W, R reals, γ a real function, $\tilde{I}$ (It) any real with $\gamma(\tilde{I}) = W/R$; StrictMono γ is Assumption 1 verbatim. The nontrivial step is `lt_div_iff0`, which moves the positive denominator across the inequality; then the claim is linear. Lean verifies the sentence *exactly*, with one hypothesis the paper leaves implicit: $\gamma > 0$. Uniqueness of $\tilde{I}$ is not needed by the row. lake build: success; check -fast: exit 0.

4 · Lean III: the Cobb-Douglas wage response — the extension that settles the trap

Claim (mine). With $\ln W = \ln c + \ln s + (1 - s) \ln K + (s - 1) \ln L$, $s = N - I^*$:

$d \ln W / ds = 1/s - \ln K + \ln L$, hence $d \ln W / dI^* = \ln(K/L) - 1/(N - I^*) > 0 \iff e^{1/(N - I^*)} < K/L$.

```
theorem cobb_douglas_log_wage_deriv (c K L s : ℝ) (hs : 0 < s) :
  HasDerivAt (fun s' : ℝ => Real.log c + Real.log s' + (1 - s') * Real.log K + (s' - 1) * Real.log L)
    (1 / s - Real.log K + Real.log L) s := by
  have h1 : HasDerivAt (fun s' : ℝ => Real.log s') (1 / s) s := by
    simp [one_div] using Real.hasDerivAt_log hs.ne'
  have h2 : HasDerivAt (fun s' : ℝ => (1 - s') * Real.log K) (-Real.log K) s := by
    have := ((hasDerivAt_id' (x := s)).const_sub 1).mul_const (Real.log K); simp using this
  have h3 : HasDerivAt (fun s' : ℝ => (s' - 1) * Real.log L) (Real.log L) s := by
    have := ((hasDerivAt_id' (x := s)).sub_const 1).mul_const (Real.log L); simp using this
  have h := (((hasDerivAt_const s (Real.log c)).add h1).add h2).add h3
  refine h.congr_deriv ?_; ring

theorem cobb_douglas_wage_rises_iff (K L s : ℝ) (hK : 0 < K) (hL : 0 < L) (hs : 0 < s) :
  0 < Real.log (K / L) - 1 / s ↔ Real.exp (1 / s) < K / L := by
  have hKL : 0 < K / L := div_pos hK hL
  rw [sub_pos]; exact Real.lt_log_iff_exp_lt hKL
```

Reading. The derivative is assembled term by term with `HasDerivAt` combinators (`Real.hasDerivAt_log`

needs only $s \neq 0$); `congr_deriv` plus `ring` matches the paper's form. The sign lemma is

`Real.lt_log_iff_exp_lt`. Since $s = N - I^*$ and $dI^* = -ds$, the wage rises with automation iff

$K/L > e^{1/(N - I^*)}$: **abundant capital**. Under Assumption 3, $K/L < (1 - s)/s < e^{1/s}$, so the printed “increases the wage when $K < \bar{K}$ ” cannot be right for this case; the numerics agree for $\hat{\sigma} \neq 1$.

5 · Where I did not believe the AI — and where I did not believe the paper

jueves, 34 de setiembre de 2036 5:37 p. m.

LI WODFLO STATION:
$$u = \mathcal{G}^T [U]_{\text{row}(u)}$$

$$y = \tilde{S} \left[\int_{\sigma_0}^u \gamma(\sigma) \frac{d\sigma}{\sigma} d_1 \right] \frac{r}{r-1}$$

2014

[illegible]

100


$$g(x) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(t) \delta(x-t) dt$$

$$\gamma(z) = \bar{\gamma}(z) \left[\gamma^{\text{tr}}(\underline{z}) \right]^{\frac{1}{2}} + (1 - \gamma^{\text{tr}}(\underline{z})) \left(\gamma^{\text{tr}}(\underline{z}) \right)^{\frac{1}{2}}$$

Un solo riferimento
per le tabelle

Journal of Management Education 32(10) 1039-1054

$$A(t) = \mathbb{E} \left[\sum_{i=1}^n \mathbb{E} \left[\frac{\partial \log \pi(\mathbf{y}_i | \mathbf{y}_{-i}, \theta)}{\partial \theta} \right] \right]$$

Acerca de este libro

1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026, 2027, 2028, 2029, 2030, 2031, 2032, 2033, 2034, 2035, 2036, 2037, 2038, 2039, 2040, 2041, 2042, 2043, 2044, 2045, 2046, 2047, 2048, 2049, 2050, 2051, 2052, 2053, 2054, 2055, 2056, 2057, 2058, 2059, 2060, 2061, 2062, 2063, 2064, 2065, 2066, 2067, 2068, 2069, 2070, 2071, 2072, 2073, 2074, 2075, 2076, 2077, 2078, 2079, 2080, 2081, 2082, 2083, 2084, 2085, 2086, 2087, 2088, 2089, 2090, 2091, 2092, 2093, 2094, 2095, 2096, 2097, 2098, 2099, 2100, 2101, 2102, 2103, 2104, 2105, 2106, 2107, 2108, 2109, 2110, 2111, 2112, 2113, 2114, 2115, 2116, 2117, 2118, 2119, 2120, 2121, 2122, 2123, 2124, 2125, 2126, 2127, 2128, 2129, 2130, 2131, 2132, 2133, 2134, 2135, 2136, 2137, 2138, 2139, 2140, 2141, 2142, 2143, 2144, 2145, 2146, 2147, 2148, 2149, 2150, 2151, 2152, 2153, 2154, 2155, 2156, 2157, 2158, 2159, 2160, 2161, 2162, 2163, 2164, 2165, 2166, 2167, 2168, 2169, 2170, 2171, 2172, 2173, 2174, 2175, 2176, 2177, 2178, 2179, 2180, 2181, 2182, 2183, 2184, 2185, 2186, 2187, 2188, 2189, 2190, 2191, 2192, 2193, 2194, 2195, 2196, 2197, 2198, 2199, 2200, 2201, 2202, 2203, 2204, 2205, 2206, 2207, 2208, 2209, 2210, 2211, 2212, 2213, 2214, 2215, 2216, 2217, 2218, 2219, 2220, 2221, 2222, 2223, 2224, 2225, 2226, 2227, 2228, 2229, 2230, 2231, 2232, 2233, 2234, 2235, 2236, 2237, 2238, 2239, 2240, 2241, 2242, 2243, 2244, 2245, 2246, 2247, 2248, 2249, 2250, 2251, 2252, 2253, 2254, 2255, 2256, 2257, 2258, 2259, 2260, 2261, 2262, 2263, 2264, 2265, 2266, 2267, 2268, 2269, 2270, 2271, 2272, 2273, 2274, 2275, 2276, 2277, 2278, 2279, 2280, 2281, 2282, 2283, 2284, 2285, 2286, 2287, 2288, 2289, 2290, 2291, 2292, 2293, 2294, 2295, 2296, 2297, 2298, 2299, 2300, 2301, 2302, 2303, 2304, 2305, 2306, 2307, 2308, 2309, 2310, 2311, 2312, 2313, 2314, 2315, 2316, 2317, 2318, 2319, 2320, 2321, 2322, 2323, 2324, 2325, 2326, 2327, 2328, 2329, 2330, 2331, 2332, 2333, 2334, 2335, 2336, 2337, 2338, 2339, 2340, 2341, 2342, 2343, 2344, 2345, 2346, 2347, 2348, 2349, 2350, 2351, 2352, 2353, 2354, 2355, 2356, 2357, 2358, 2359, 2360, 2361, 2362, 2363, 2364, 2365, 2366, 2367, 2368, 2369, 2370, 2371, 2372, 2373, 2374, 2375, 2376, 2377, 2378, 2379, 2380, 2381, 2382, 2383, 2384, 2385, 2386, 2387, 2388, 2389, 2390, 2391, 2392, 2393, 2394, 2395, 2396, 2397, 2398, 2399, 2400, 2401, 2402, 2403, 2404, 2405, 2406, 2407, 2408, 2409, 2410, 2411, 2412, 2413, 2414, 2415, 2416, 2417, 2418, 2419, 2420, 2421, 2422, 2423, 2424, 2425, 2426, 2427, 2428, 2429, 2430, 2431, 2432, 2433, 2434, 2435, 2436, 2437, 2438, 2439, 2440, 2441, 2442, 2443, 2444, 2445, 2446, 2447, 2448, 2449, 2450, 2451, 2452, 2453, 2454, 2455, 2456, 2457, 2458, 2459, 2460, 2461, 2462, 2463, 2464, 2465, 2466, 2467, 2468, 2469, 2470, 2471, 2472, 2473, 2474, 2475, 2476, 2477, 2478, 2479, 2480, 2481, 2482, 2483, 2484, 2485, 2486, 2487, 2488, 2489, 2490, 2491, 2492, 2493, 2494, 2495, 2496, 2497, 2498, 2499, 2500, 2501, 2502, 2503, 2504, 2505, 2506, 2507, 2508, 2509, 2510, 2511, 2512, 2513, 2514, 2515, 2516, 2517, 2518, 2519, 2520, 2521, 2522, 2523, 2524, 2525, 2526, 2527, 2528, 2529, 2530, 2531, 2532, 2533, 2534, 2535, 2536, 2537, 2538, 2539, 2540, 2541, 2542, 2543, 2544, 2545, 2546, 2547, 2548, 2549, 2550, 2551, 2552, 2553, 2554, 2555, 2556, 2557, 2558, 2559, 2560, 2561, 2562, 2563, 2564, 2565, 2566, 2567, 2568, 2569, 2570, 2571, 2572, 2573, 2574, 2575, 2576, 2577, 2578, 2579, 2580, 2581, 2582, 2583, 2584, 2585, 2586, 2587, 2588, 2589, 2590, 2591, 2592, 2593, 2594, 2595, 2596, 2597, 2598, 2599, 2600, 2601, 2602, 2603, 2604, 2605, 2606, 2607, 2608, 2609, 2610, 2611, 2612, 2613, 2614, 2615, 2616, 2617, 2618, 2619, 2620, 2621, 2622, 2623, 2624, 2625, 2626, 2627, 2628, 2629, 2630, 2631, 2632, 2633, 2634, 2635, 2636, 2637, 2638, 2639, 2640, 2641, 2642, 2643, 2644, 2645, 2646, 2647, 2648, 2649, 2650, 2651, 2652, 2653, 2654, 2655, 2656, 2657, 2658, 2659, 2660, 2661, 2662, 2663, 2664, 2665, 2666, 2667, 2668, 2669, 2670, 2671, 2672, 2673, 2674, 2675, 2676, 2677, 2678, 2679, 26

E1 *Homomorphisms from L to L are derivations*

 (10^{-10})

$$V(L_1 L_2) = \frac{(L_1 e^{-\tau(L_1)})^{1-\theta} - 1}{1-\theta}$$

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$$\eta \rightarrow 0 \quad \xi = 1$$

$\eta \rightarrow 0 \quad \xi = 1$

How. Col. 1/10/18. 1/10/18. 1/10/18.

Case No. 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405, 406, 407, 408, 409, 410, 411, 412, 413, 414, 415, 416, 417, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 431, 432, 433, 434, 435, 436, 437, 438, 439, 440, 441, 442, 443, 444, 445, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455, 456, 457, 458, 459, 460, 461, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471, 472, 473, 474, 475, 476, 477, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492, 493, 494, 495, 496, 497, 498, 499, 500, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 511, 512, 513, 514, 515, 516, 517, 518, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533, 534, 535, 536, 537, 538, 539, 540, 541, 542, 543, 544, 545, 546, 547, 548, 549, 550, 551, 552, 553, 554, 555, 556, 557, 558, 559, 560, 561, 562, 563, 564, 565, 566, 567, 568, 569, 570, 571, 572, 573, 574, 575, 576, 577, 578, 579, 580, 581, 582, 583, 584, 585, 586, 587, 588, 589, 590, 591, 592, 593, 594, 595, 596, 597, 598, 599, 600, 601, 602, 603, 604, 605, 606, 607, 608, 609, 610, 611, 612, 613, 614, 615, 616, 617, 618, 619, 620, 621, 622, 623, 624, 625, 626, 627, 628, 629, 630, 631, 632, 633, 634, 635, 636, 637, 638, 639, 640, 641, 642, 643, 644, 645, 646, 647, 648, 649, 650, 651, 652, 653, 654, 655, 656, 657, 658, 659, 660, 661, 662, 663, 664, 665, 666, 667, 668, 669, 670, 671, 672, 673, 674, 675, 676, 677, 678, 679, 680, 681, 682, 683, 684, 685, 686, 687, 688, 689, 690, 691, 692, 693, 694, 695, 696, 697, 698, 699, 700, 701, 702, 703, 704, 705, 706, 707, 708, 709, 710, 711, 712, 713, 714, 715, 716, 717, 718, 719, 720, 721, 722, 723, 724, 725, 726, 727, 728, 729, 730, 731, 732, 733, 734, 735, 736, 737, 738, 739, 740, 741, 742, 743, 744, 745, 746, 747, 748, 749, 750, 751, 752, 753, 754, 755, 756, 757, 758, 759, 760, 761, 762, 763, 764, 765, 766, 767, 768, 769, 770, 771, 772, 773, 774, 775, 776, 777, 778, 779, 780, 781, 782, 783, 784, 785, 786, 787, 788, 789, 790, 791, 792, 793, 794, 795, 796, 797, 798, 799, 800, 801, 802, 803, 804, 805, 806, 807, 808, 809, 810, 811, 812, 813, 814, 815, 816, 817, 818, 819, 820, 821, 822, 823, 824, 825, 826, 827, 828, 829, 830, 831, 832, 833, 834, 835, 836, 837, 838, 839, 840, 841, 842, 843, 844, 845, 846, 847, 848, 849, 850, 851, 852, 853, 854, 855, 856, 857, 858, 859, 860, 861, 862, 863, 864, 865, 866, 867, 868, 869, 870, 871, 872, 873, 874, 875, 876, 877, 878, 879, 880, 881, 882, 883, 884, 885, 886, 887, 888, 889, 890, 891, 892, 893, 894, 895, 896, 897, 898, 899, 900, 901, 902, 903, 904, 905, 906, 907, 908, 909, 910, 911, 912, 913, 914, 915, 916, 917, 918,

Hand derivation, part 1 of 3: the task CES and the unit-cost minimisation behind $p(i)$. Parts 2–3 (the threshold, the two

Claim 1 (the flat answer): “automation lowers wages and the labor share.” Props. 2–3: the labor share always falls when technology binds; the wage rises iff productivity beats displacement, and does not move in the cost-minimising regime. **Verdict: half right.**

Claim 2 (my first numerics): I wrote $W/R = \omega K/L$; Prop. 2's $d \ln(W/R)/dI = d \ln \omega/dI$ then failed by $(1 - \varepsilon_L)$. The paper's $\omega = W/(RK)$ gives $W/R = \omega K$; fixed, the numerical derivative matches Prop. 3's formula to four decimals. **Verdict: I was wrong; the check caught it.**

Claim 3 (the paper): “there exists $\bar{K} > \underline{K}$ such that an increase in I increases the wage when $K < \bar{K}$ and reduces it when $K > \bar{K}$.” By hand and in Lean, Cobb-Douglas gives $d \ln W / dI^* = \ln(K/L) - 1/(N - I^*)$: the wage rises for large K , so with $\bar{K} > \underline{K}$ it always falls inside Assumption 3; the $\hat{\sigma} = 0.6$ numerics and the paper’s own intuition (cost savings $W/\gamma(I^*) - R$) agree. **Verdict: the two inequalities are printed the wrong way round** (AER text not checked).

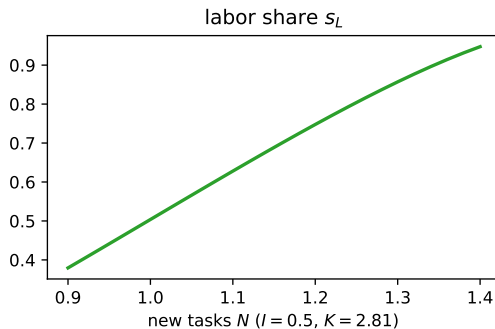
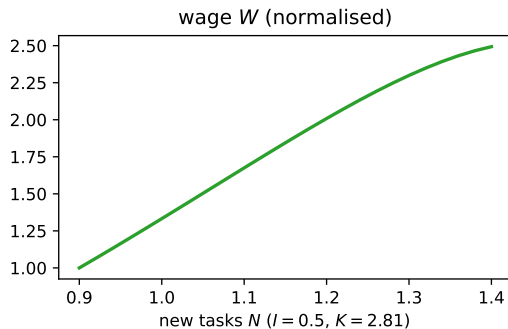
Claim 4 (Lean pushed back): the productivity term divides by $1 - \hat{\sigma}$; the sign proof needs $\hat{\sigma} < 1$ and $\hat{\sigma} > 1$ separately and excludes $\hat{\sigma} = 1$ — the Cobb-Douglas limit the paper takes silently.

5 · What to carry away

- ▶ **Two forces, not one.** Automation *displaces* (fewer tasks for labor: W/R , s_L , L fall) but also raises productivity; new tasks *reinstates* labor. The race is between I and N .
- ▶ **The wage condition.** Automation raises the wage iff the cost saving on the marginal task, $W/\gamma(I^*) - R$, outweighs $(1 - s_L)\Lambda_I/(\hat{\sigma} + \varepsilon_L)$ — i.e. when capital is abundant (Cobb-Douglas: $K/L > e^{1/(N-I^*)}$). Under the paper's Assumption 3 it falls.
- ▶ **Regime matters.** If technology does not bind ($I^* = \tilde{I} < I$), better automation technology changes nothing; only N moves prices.
- ▶ **Lean.** Ten rows proved exactly from the displays; the equilibrium map and the general $\bar{K}$ are the declared boundary; the extension proves the closed-form threshold and exposes the reversed inequality.
- ▶ **Versions.** NBER June 2017 (course PDF) read and pinned; AER 2018 reverses the title; the page numbers here are the PDF's.

B1 · New tasks: W and s_L always rise (reinstatement)

New tasks (reinstatement): wage and labor share always rise (Propositions 2-3)



B2 · Hand derivation (hand/hand-derivation.pdf): the static model, the threshold, the task demand

Acemoglu y Restrepo (2018)

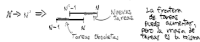
January 24 de septiembre de 2018 3:37 p.m.

EL MODELO ESTÁTICO

Entonces en una economía compuesta de tareas. La producción final Y depende de tareas. La productividad de una tarea i es y_i de donde viene $y_i(u_i)$ entonces:

$$Y = \sum_{i=1}^N y_i(u_i) \frac{1}{\sigma} d_i \frac{1}{\sigma}$$

Las $i^{\text{ésimas}}$ tareas siempre están en un intervalo unitario: $i \in [0, 1]$. Donde U representa la frontera de tareas.



Se define a " U " como la frontera de automatización.



La decisión de automatizar o no dependerá de los costos de las tareas. Si decir, no todo $i \in \mathbb{R}$ es automatizable. Para las tareas no automatizables, su función de producción es:

$$y_i(u_i) = \frac{1}{\sigma} \left[\eta^{\frac{\sigma}{1-\sigma}} \frac{1}{\sigma} + (1-\eta) \frac{1}{\sigma} \right] \frac{1}{\sigma} \frac{1}{\sigma}$$

Un input intermediario para la tarea i Probabilidad de éxito del trabajo Tercer Estándar

Assumption 1: $y_i(u_i) > 0$; Mas avanzando la tarea $\rightarrow$ $y_i(u_i)$

Con tareas automatizables, la producción es:

$$y_i(u_i) = \frac{1}{\sigma} \left[\eta^{\frac{\sigma}{1-\sigma}} \frac{1}{\sigma} + (1-\eta) \frac{1}{\sigma} \right] \frac{1}{\sigma} \frac{1}{\sigma}$$

Reemplaza R como el costo de trabajo de la tarea con productividad normalizada a 1.

Como $y_i(u_i) > 0$ El costo de trabajo para bajo i . El trabajo a partes para alto i .

El hogar tiene las siguientes preferencias:

$$U(C, L) = \frac{C^{1-\theta} (1-L)^{\theta}}{1-\theta}$$

C: Consumo; L: Trabajo; $U(L)$: Desutilidad por L

RESUMEN

Lemma 2: Se necesita 1 de entre 2 condiciones.

$$\eta \rightarrow 0 \quad \sigma = 1$$

Para tener soluciones cerradas.

Para cualquier tarea i y su precio de venta será igual a su costo unitario en competencia perfecta. Como se sabe, el precio con $R = 1$ cuando $\sigma = 1$ y $\theta = 1$ consideramos el mismo caso.

Costo del trabajo $\rightarrow$ Salarios

$$\text{Costo del Trabajo} = \frac{w}{Y} \leftarrow \text{Salarios por unidad efectiva}$$

$$\text{Como } Y'(L) > 0 \rightarrow \eta \rightarrow \frac{Y'(L)}{Y(L)}$$

Por lo tanto, el costo unitario R , por lo que la firma usa:

$$\min \{ R, \frac{w}{Y(L)} \}; \quad i \in I$$

Además, asumiendo $\sigma = 1$, entonces la función $Y(L)$:

$$Y(L) = \frac{1}{\sigma} \left[\eta^{\frac{\sigma}{1-\sigma}} \frac{1}{\sigma} + (1-\eta) \frac{1}{\sigma} \right] \frac{1}{\sigma} \frac{1}{\sigma}$$

$$Y(L) = \frac{1}{\sigma} \eta^{\frac{\sigma}{1-\sigma}} \frac{1}{\sigma}$$

Recordemos que $q(u)$ es el valor incremental agregado para i y Y a la tarea Y . La firma resuelve el problema de minimización del costo, definimos $r_x = w/Y(u)$:

$$\min_{x, l} \quad \psi x + r_x x \quad \text{s.t.} \quad Y = 1 = \frac{1}{\sigma} \eta^{\frac{\sigma}{1-\sigma}} \frac{1}{\sigma}$$

de fijar en l para obtener el óptimo unitario

$$L = \psi x + r_x x + \lambda [1 - \frac{1}{\sigma} \eta^{\frac{\sigma}{1-\sigma}} \frac{1}{\sigma}]$$

$$\frac{\partial L}{\partial x} = \psi + \lambda \frac{1}{\sigma} \eta^{\frac{\sigma}{1-\sigma}} \frac{1}{\sigma} = 0 \quad \frac{\partial L}{\partial \lambda} = 1 - \frac{1}{\sigma} \eta^{\frac{\sigma}{1-\sigma}} \frac{1}{\sigma} = 0$$

$$\frac{\partial L}{\partial x} = \psi + \lambda \frac{1}{\sigma} \eta^{\frac{\sigma}{1-\sigma}} \frac{1}{\sigma} = 0 \quad \frac{\partial L}{\partial \lambda} = 1 - \frac{1}{\sigma} \eta^{\frac{\sigma}{1-\sigma}} \frac{1}{\sigma} = 0$$

$$\rightarrow x = \frac{1-\eta}{\eta} \frac{1}{\sigma} \frac{1}{\sigma} \quad \text{de donde se obtiene en la multiplicación}$$

$$\rightarrow 1 = \frac{1}{\sigma} \eta^{\frac{\sigma}{1-\sigma}} \frac{1}{\sigma} \frac{1}{\sigma} = \frac{1}{\sigma} \left[\frac{1-\eta}{\eta} \frac{1}{\sigma} \right] \frac{1}{\sigma} \frac{1}{\sigma}$$

$$\rightarrow \frac{1}{\sigma} = \frac{1}{\sigma} \left[\frac{1-\eta}{\eta} \frac{1}{\sigma} \right] \frac{1}{\sigma} \frac{1}{\sigma} \quad \text{Reemplazando (1) en (2)}$$

$$\rightarrow x^* = \frac{1-\eta}{\eta} \frac{1}{\sigma} \frac{1}{\sigma} = \left[\frac{1-\eta}{\eta} \frac{1}{\sigma} \right] \frac{1}{\sigma} \frac{1}{\sigma}$$

$$\rightarrow x^* = \frac{1}{\sigma} \left[\frac{1-\eta}{\eta} \frac{1}{\sigma} \right] \frac{1}{\sigma}$$

$$\min C(L) = \psi x^* + r_x x^* = \frac{1}{\sigma} \left[\frac{1-\eta}{\eta} \frac{1}{\sigma} \right] \frac{1}{\sigma} \frac{1}{\sigma} + r_x \frac{1}{\sigma} \left[\frac{1-\eta}{\eta} \frac{1}{\sigma} \right] \frac{1}{\sigma}$$

$$C(L) = \frac{1}{\sigma} \left[\frac{1-\eta}{\eta} \frac{1}{\sigma} \right] \frac{1}{\sigma} \frac{1}{\sigma} + r_x \frac{1}{\sigma} \left[\frac{1-\eta}{\eta} \frac{1}{\sigma} \right] \frac{1}{\sigma}$$

$$C(L) = \frac{1}{\sigma} \left[\frac{1-\eta}{\eta} \frac{1}{\sigma} \right] \frac{1}{\sigma} \frac{1}{\sigma} + r_x \frac{1}{\sigma} \left[\frac{1-\eta}{\eta} \frac{1}{\sigma} \right] \frac{1}{\sigma}$$

$$\text{Así los autores usan:}$$

$$\frac{1}{\sigma} = \psi^{\frac{1}{1-\theta}} (1-\eta)^{\frac{\theta}{1-\theta}} \frac{1}{\sigma}$$

Entonces:

$$C(L) = \frac{1}{\sigma} \left[\frac{1-\eta}{\eta} \frac{1}{\sigma} \right] \frac{1}{\sigma} \frac{1}{\sigma}$$

$$C(L) = r_x \frac{1}{\sigma} = \left(\frac{1}{\sigma} \right) \frac{1}{\sigma}$$

Ojo que este es el caso cuando $x = Y(L) \frac{1}{\sigma}$, si evaluáramos con $x = Y(L)$, tendríamos:

$$C(L) = \frac{1}{\sigma} \left[\frac{1-\eta}{\eta} \frac{1}{\sigma} \right] \frac{1}{\sigma} \frac{1}{\sigma}$$

En conclusión:

B3 · Lean IV: Proposition 3's productivity effect on both sides of $\hat{\sigma} = 1$

```
def paper_prop3_productivity_effect_positiveSpec : Prop :=
  ∀ Wg R σh : ℝ, 0 < R → R < Wg → σh ≠ 1 →
    0 < (Wg ^ (1 - σh) - R ^ (1 - σh)) / (1 - σh)

theorem paper_prop3_productivity_effect_positive : paper_prop3_productivity_effect_positiveSpec := by
  intro Wg R σh hR hlt hne
  rcases lt_or_gt_of_ne hne with hlt1 | hgt1
  · have hexp : 0 < 1 - σh := by linarith
    have h : R ^ (1 - σh) < Wg ^ (1 - σh) := Real.rpow_lt_rpow hR.le hlt hexp
    exact div_pos (by linarith) hexp
  · have hexp : 1 - σh < 0 := by linarith
    have h : Wg ^ (1 - σh) < R ^ (1 - σh) := Real.rpow_lt_rpow_of_neg hR hlt hexp
    exact div_pos_of_neg_of_neg (by linarith) hexp

def paper_prop3_wage_decompositionSpec : Prop :=
  ∀ P sL ΔI σh εL dI : ℝ,
    0 < P - (1 - sL) * ((1 / (σh + εL)) * ΔI * dI) ↔
      (1 - sL) * ((1 / (σh + εL)) * ΔI * dI) < P
```

Reading. Wg is $W/\gamma(I^*)$ and R the rental rate; the hypothesis $R < Wg$ is the constrained case of Proposition 3. For $\hat{\sigma} < 1$ the real power is increasing and both numerator and denominator are positive; for $\hat{\sigma} > 1$ both are negative. The decomposition row states the wage sign as an equivalence over the paper's display: productivity effect P against displacement.